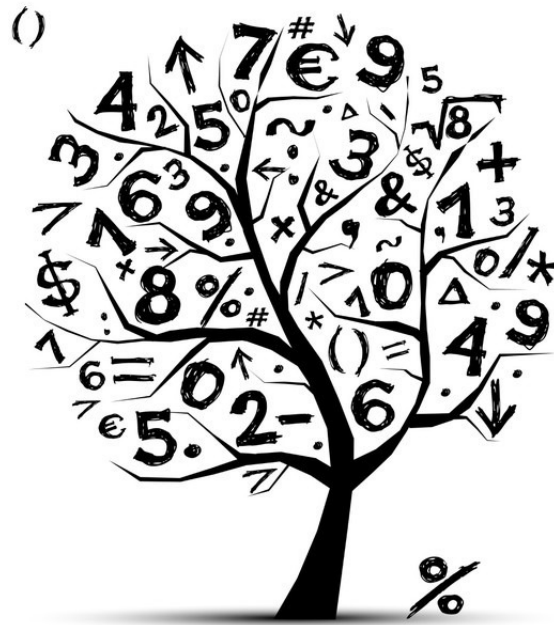


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Exercises: sums



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Exercise 1

Prove, using the properties of the sums, that for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n k^2 = \frac{2n^3 + 3n^2 + n}{6} \quad (1)$$

$= \frac{n(n+1)(2n+1)}{6}$

(sum of the squares of the first n natural numbers without zero).

We start by writing $\sum_{k=0}^n (k+1)^3$ in two different ways:

$$\begin{aligned} 1) \quad \sum_{k=0}^n (k+1)^3 &= \sum_{k=0}^n (k^3 + 3k^2 + 3k + 1) \\ &= \left(\sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + 3 \sum_{k=1}^n k + \sum_{k=1}^n 1 \right) + 1 \\ &= \sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + \frac{3(n^2 + n)}{2} + n + 1 \end{aligned}$$

$$2) \quad \sum_{k=0}^n (k+1)^3 = \sum_{k=1}^{n+1} k^3 = \sum_{k=1}^n k^3 + (n+1)^3$$

(in 1) the term with $k = 0$, equal to 1, has been separated from the others). Equating the expressions and canceling the sum with k^3 we get:

$$3 \sum_{k=1}^n k^2 + \frac{3(n^2 + n)}{2} + n + 1 = (n+1)^3$$

Isolating the sum with k^2 we get:

$$\begin{aligned} 3 \sum_{k=1}^n k^2 &= (n+1)^3 - \frac{3(n^2 + n)}{2} - n - 1 \\ &= n^3 + 3n^2 + 3n + 1 - \frac{3n^2 + 3n}{2} - n - 1 \\ &= \frac{2n^3 + 3n^2 + n}{2} \end{aligned}$$

Then we have:

$$\sum_{k=1}^n k^2 = \frac{2n^3 + 3n^2 + n}{6}$$

Exercise 2

Compute the sum of the squares of the first 10, 100 and 1000 natural numbers (without zero).

Using formula (1), the sum of squares of the first 10 natural numbers (without zero) is:

$$\sum_{k=1}^{10} k^2 = \frac{2 \cdot 10^3 + 3 \cdot 10^2 + 10}{6} = 385$$

The sum of squares of the first 100 natural numbers (without zero) is:

$$\sum_{k=1}^{100} k^2 = \frac{2 \cdot 100^3 + 3 \cdot 100^2 + 100}{6} = 338.350$$

The sum of squares of the first 1000 natural numbers (without zero) is:

$$\sum_{k=1}^{1000} k^2 = \frac{2 \cdot 1000^3 + 3 \cdot 1000^2 + 1000}{6} = 333.833.500$$

Exercise 3

Prove, using the properties of the sums, that for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n k^3 = \frac{n^4 + 2n^3 + n^2}{4} \quad (2)$$

$$= \frac{n^2(n+1)^2}{4} = \frac{(n^2+n)^2}{4}$$

(sum of the cubes of the first n natural numbers without zero).

We start by writing $\sum_{k=0}^n (k+1)^4$ in two different ways, using formula (1) for the sum of the squares:

$$\begin{aligned}
 1) \quad \sum_{k=0}^n (k+1)^4 &= \sum_{k=0}^n (k^4 + 4k^3 + 6k^2 + 4k + 1) \\
 &= \left(\sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 6 \sum_{k=1}^n k^2 + 4 \sum_{k=1}^n k + \sum_{k=1}^n 1 \right) + 1 \\
 &= \sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 2n^3 + 3n^2 + n + 2n^2 + 2n + n + 1 \\
 &= \sum_{k=1}^n k^4 + 4 \sum_{k=1}^n k^3 + 2n^3 + 5n^2 + 4n + 1 \\
 2) \quad \sum_{k=0}^n (k+1)^4 &= \sum_{k=1}^{n+1} k^4 = \sum_{k=1}^n k^4 + (n+1)^4
 \end{aligned}$$

Equating the expressions and canceling the sum with k^4 we get:

$$4 \sum_{k=1}^n k^3 + 2n^3 + 5n^2 + 4n + 1 = (n+1)^4$$

Isolating the sum with k^3 we get:

$$4 \sum_{k=1}^n k^3 = n^4 + 4n^3 + 6n^2 + 4n + 1 - 2n^3 - 5n^2 - 4n - 1 = n^4 + 2n^3 + n^2$$

Then we have:

$$\sum_{k=1}^n k^3 = \frac{n^4 + 2n^3 + n^2}{4}$$

Exercise 4

Compute the sum of the cubes of the first 10, 100 and 1000 natural numbers (without zero).

Using formula (2), the sum of cubes of the first 10 natural numbers (without zero) is:

$$\sum_{k=1}^{10} k^3 = \frac{10^4 + 2 \cdot 10^3 + 10^2}{4} = 3025$$

The sum of cubes of the first 100 natural numbers (without zero) is:

$$\sum_{k=1}^{100} k^3 = \frac{100^4 + 2 \cdot 100^3 + 100^2}{4} = 25.502.500$$

The sum of cubes of the first 1000 natural numbers (without zero) is:

$$\sum_{k=1}^{1000} k^3 = \frac{1000^4 + 2 \cdot 1000^3 + 1000^2}{4} = 250.500.250.000$$

Exercise 5

Prove that the sum of the cubes of the first n natural numbers (without zero) equals the square of the sum of the first n natural numbers (without zero):

$$\sum_{k=1}^n k^3 = \left(\sum_{k=1}^n k \right)^2$$

Using formula (2) and the formula $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, we have:

$$\sum_{k=1}^n k^3 = \frac{n^4 + 2n^3 + n^2}{4} = \frac{n^2(n^2 + 2n + 1)}{4} = \frac{n^2(n+1)^2}{4} = \left(\frac{n(n+1)}{2} \right)^2 = \left(\sum_{k=1}^n k \right)^2$$

Exercise 6

Using the index translation, rewrite the following sums so that the sum index starts from 1, and compute them:

$$\text{a) } \sum_{j=3}^{12} (j-2)^2, \quad \text{b) } \sum_{j=0}^{n-1} (j+1), \quad \text{c) } \sum_{j=0}^{49} (2j+1)$$

- a) With the index translation by $m = 2$ (setting $i = j - 2$, when j goes from 3 to 12, i goes from 1 to 10) and formula (1):

$$\sum_{j=3}^{12} (j-2)^2 = \sum_{i=1}^{10} i^2 = \frac{2 \cdot 10^3 + 3 \cdot 10^2 + 10}{6} = 385$$

- b) With the index translation by $m = 1$ (setting $i = j + 1$):

$$\sum_{j=0}^{n-1} (j+1) = \sum_{i=1}^n i = \frac{n^2 + n}{2}$$

- c) With the index translation by $m = 1$ (setting $i = j + 1$, so that $2j + 1 = 2i - 1$), this is the sum of the first 50 odd numbers:

$$\sum_{j=0}^{49} (2j+1) = \sum_{i=1}^{50} (2i-1) = 50^2 = 2500$$

Exercise 7

Find a closed formula for the sum

$$\sum_{j=1}^n (3j^2 - 2j + 5)$$

and use it to compute the value for $n = 10$.

Using the union of sums, the product by a constant and the sum with constant term, we have:

$$\begin{aligned} \sum_{j=1}^n (3j^2 - 2j + 5) &= 3 \sum_{j=1}^n j^2 - 2 \sum_{j=1}^n j + \sum_{j=1}^n 5 \\ &= 3 \frac{2n^3 + 3n^2 + n}{6} - 2 \frac{n^2 + n}{2} + 5n \\ &= n^3 + \frac{3}{2}n^2 + \frac{1}{2}n - n^2 - n + 5n = \frac{2n^3 + n^2 + 9n}{2} \end{aligned}$$

For $n = 10$ we get:

$$\sum_{j=1}^{10} (3j^2 - 2j + 5) = \frac{2000 + 100 + 90}{2} = 1095$$

As a check: $3 \cdot 385 - 2 \cdot 55 + 5 \cdot 10 = 1155 - 110 + 50 = 1095$.

Exercise 8

Using the decomposition property, compute:

$$\text{a) } \sum_{j=11}^{20} j^2, \quad \text{b) } \sum_{j=n+1}^{2n} j \quad (n \geq 1)$$

a) From the decomposition $\sum_{j=1}^{20} j^2 = \sum_{j=1}^{10} j^2 + \sum_{j=11}^{20} j^2$ and formula (1):

$$\sum_{j=11}^{20} j^2 = \sum_{j=1}^{20} j^2 - \sum_{j=1}^{10} j^2 = \frac{2 \cdot 20^3 + 3 \cdot 20^2 + 20}{6} - 385 = 2870 - 385 = 2485$$

b) From the decomposition $\sum_{j=1}^{2n} j = \sum_{j=1}^n j + \sum_{j=n+1}^{2n} j$:

$$\sum_{j=n+1}^{2n} j = \frac{(2n)^2 + 2n}{2} - \frac{n^2 + n}{2} = \frac{4n^2 + 2n - n^2 - n}{2} = \frac{3n^2 + n}{2}$$

Exercise 9

a) Prove that for every sequence of real numbers $a_1, a_2, \dots, a_{n+1}$ we have (**telescoping sum**):

$$\sum_{j=1}^n (a_{j+1} - a_j) = a_{n+1} - a_1$$

b) Using a), compute $\sum_{j=1}^n \frac{1}{j(j+1)}$ and its value for $n = 99$.

- a) Using the union of sums, the index translation and the decomposition, we have:

$$\begin{aligned}\sum_{j=1}^n (a_{j+1} - a_j) &= \sum_{j=1}^n a_{j+1} - \sum_{j=1}^n a_j = \sum_{j=2}^{n+1} a_j - \sum_{j=1}^n a_j \\ &= \left(\sum_{j=2}^n a_j + a_{n+1} \right) - \left(a_1 + \sum_{j=2}^n a_j \right) = a_{n+1} - a_1\end{aligned}$$

- b) Since

$$\frac{1}{j(j+1)} = \frac{(j+1) - j}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}$$

we can apply a) with $a_j = -\frac{1}{j}$:

$$\sum_{j=1}^n \frac{1}{j(j+1)} = \sum_{j=1}^n \left(-\frac{1}{j+1} - \left(-\frac{1}{j} \right) \right) = -\frac{1}{n+1} + 1 = \frac{n}{n+1}$$

For $n = 99$ we get $\frac{99}{100}$.

Exercise 10

Compute $\sum_{j=1}^n ((j+1)^2 - j^2)$ in two different ways and deduce again the formula

$$\sum_{j=1}^n j = \frac{n^2 + n}{2}$$

First way: it is a telescoping sum (Exercise 9) with $a_j = j^2$, then:

$$\sum_{j=1}^n ((j+1)^2 - j^2) = (n+1)^2 - 1 = n^2 + 2n$$

Second way: since $(j+1)^2 - j^2 = 2j + 1$, we have:

$$\sum_{j=1}^n ((j+1)^2 - j^2) = \sum_{j=1}^n (2j + 1) = 2 \sum_{j=1}^n j + n$$

Equating the two expressions:

$$2 \sum_{j=1}^n j + n = n^2 + 2n \quad \text{that is} \quad \sum_{j=1}^n j = \frac{n^2 + n}{2}$$

Exercise 11

A **double sum** $\sum_{i=1}^n \sum_{j=1}^m a_{ij}$ is the sum for i from 1 to n of the inner sums $\sum_{j=1}^m a_{ij}$. Compute, for every pair of natural numbers $n, m \geq 1$:

$$\text{a) } \sum_{i=1}^n \sum_{j=1}^m (i+j), \quad \text{b) } \sum_{i=1}^n \sum_{j=1}^n i j$$

and the value of a) for $n = 3$ and $m = 4$.

a) For the inner sum (i is constant with respect to j):

$$\sum_{j=1}^m (i+j) = \sum_{j=1}^m i + \sum_{j=1}^m j = m i + \frac{m^2 + m}{2}$$

then

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m (i+j) &= \sum_{i=1}^n \left(m i + \frac{m^2 + m}{2} \right) = m \frac{n^2 + n}{2} + n \frac{m^2 + m}{2} \\ &= \frac{n m (n+1) + n m (m+1)}{2} = \frac{n m (n+m+2)}{2} \end{aligned}$$

For $n = 3$ and $m = 4$ we get $\frac{3 \cdot 4 \cdot 9}{2} = 54$.

b) Using twice the product by a constant:

$$\sum_{i=1}^n \sum_{j=1}^n i j = \sum_{i=1}^n \left(i \sum_{j=1}^n j \right) = \left(\sum_{j=1}^n j \right) \sum_{i=1}^n i = \left(\frac{n^2 + n}{2} \right)^2$$

Exercise 12

Compute, for every natural number $n \geq 1$, the following double sums in which the upper index of the inner sum depends on i :

$$\text{a) } \sum_{i=1}^n \sum_{j=1}^i 1, \quad \text{b) } \sum_{i=1}^n \sum_{j=1}^i j$$

and the value of b) for $n = 10$.

a) Since $\sum_{j=1}^i 1 = i$, we have:

$$\sum_{i=1}^n \sum_{j=1}^i 1 = \sum_{i=1}^n i = \frac{n^2 + n}{2}$$

b) Since $\sum_{j=1}^i j = \frac{i^2 + i}{2}$, using formula (1) we have:

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^i j &= \sum_{i=1}^n \frac{i^2 + i}{2} = \frac{1}{2} \left(\sum_{i=1}^n i^2 + \sum_{i=1}^n i \right) = \frac{1}{2} \left(\frac{2n^3 + 3n^2 + n}{6} + \frac{n^2 + n}{2} \right) \\ &= \frac{2n^3 + 6n^2 + 4n}{12} = \frac{n(n+1)(n+2)}{6} \end{aligned}$$

For $n = 10$ we get $\frac{10 \cdot 11 \cdot 12}{6} = 220$.